Analytics Assignment 2

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1 Introduction

The aim of this assignment is to answer the question "What clinical and demographic factors predict the development of diabetes in Pima Indian women, and how accurately can we classify diabetes status using these measurements?" This population had one of the highest rates of diabetes in the world, the data was collected by the US National Institute of Diabetes and Digestive and Kidney Diseases to from what is now a publicly available dataset [1].

1.1 Data Cleaning

The first step before carrying out any analysis was to clean the data, replacing nonsensical values with "N/A".

1.2 Summary Statistics

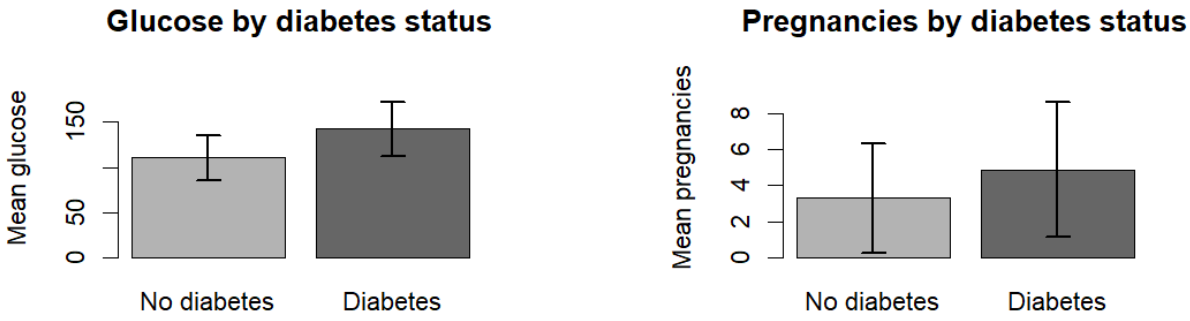
Next, summary statistics (mean ± standard deviation (SD)) were calculated for all variables, stratified by diabetes status (**Table 1**).

Table 1: Summary statistics (Mean ± SD) stratified by diabetes status.

Variable	Neg (Mean ± SD)	Pos (Mean ± SD)
Age	28.35 ± 8.99	35.94 ± 10.63
Glucose	111.43 ± 24.64	145.19 ± 29.84
Insulin	130.86 ± 102.63	206.85 ± 132.70
Mass	31.75 ± 6.79	35.78 ± 6.73
Pedigree	0.472 ± 0.299	0.626 ± 0.406
Pregnant	2.72 ± 2.62	4.47 ± 3.92
Pressure	68.97 ± 11.89	74.08 ± 13.02
Triceps	27.25 ± 10.43	32.96 ± 9.64

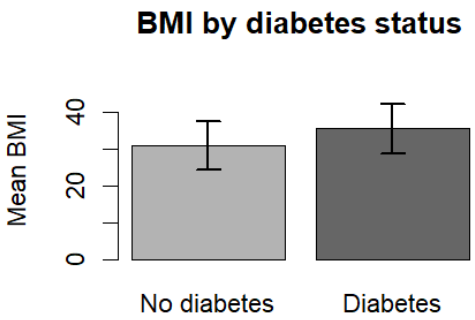
1.3 Visual Comparison of key predictors

Glucose, BMI, Pregnancy and Age were highlighted as potential key variables based on their biological significance and from the apparent difference in the output of the mean and summary statistics calculated in (**Table 1**). However, after plotting the variables (**Figure 1**) there was not one standout, predictive variable. This reinforces the need for logistic regression to create a robust model.

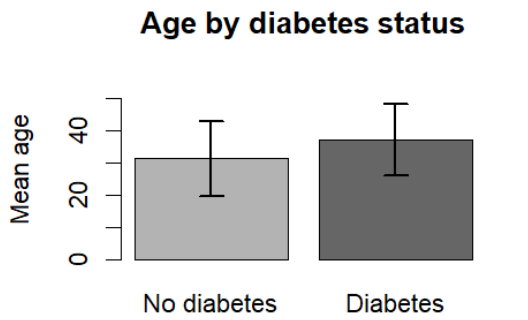


(a) Glucose

(b) Pregnancies



(c) BMI



(d) Age

Figure 1: Comparison of key predictors stratified by diabetes status (mean ± SD).

2 Designing the Optimum Model

2.1 Diabetes Status Predictive Model

Firstly, the factor levels in diabetes, positive and negative, were converted into binary. Then a model containing all the variables was created. Pregnancy, blood pressure, triceps, and insulin were not significant ($P < 0.05$) and were removed. The Akaike Information Criterion (AIC) was 362.02. The final model included glucose ($P = 3.24 \times 10^{-11}$), mass ($p = 0.0099$), pedigree (family history) ($p = 0.0076$). Age while borderline ($p = 0.65$), was included on an established biology basis as a predictor [2]. This was arguably justified as it was significant in the reduced model, however that could also be attributed to more complex age related interactions, not accounted for by the simpler logistic model. The AIC of the final model was 357.23, a slight improvement over the first model. The equation of the final model is shown in (**Equation 1**).

$$\begin{aligned}\log\left(\frac{P(\text{diabetes}=1)}{1-P(\text{diabetes}=1)}\right) &= -10.09 + 0.036\,\text{glucose} + 0.074\,\text{mass} \\ &\quad + 1.09\,\text{pedigree} + 0.053\,\text{age}\end{aligned} \tag{1}$$

Table 2: Logistic regression model predicting diabetes

Variable	Estimate	Std. Error	z value	p-value
Intercept	-10.092018	1.080251	-9.342	$< 2 \times 10^{-16}$
Glucose	0.036189	0.004982	7.264	3.76×10^{-13}
Mass (BMI)	0.074449	0.020267	3.673	0.000239
Pedigree	1.087129	0.419408	2.592	0.009541
Age	0.053012	0.013439	3.945	8.00×10^{-5}

2.2 Interpretation of model descriptors

All odds ratios in the final model had confidence intervals excluding 1, indicating statistical significance. Pedigree had the highest odds ratio (OR = 2.966) (**Table 3**), indicating that for each unit increase in the patient's "Diabetes Pedigree Function", the odds of being positive for diabetes are 2.966 times higher. This highlights the significant influence genetics has in the development of diabetes. Additionally, this data is from a native population which historically have been socially segregated. It could be interpreted that the reduction in genetic diversity, with the presence of founder mutations, has been a major contributor to the high prevalence of diabetes in the population. The next highest odds ratio was mass (OR=1.077), which stands for body mass index (BMI). This is a commonly used measure for obesity, which is a well-established risk factor for diabetes [3]. Glucose was associated with 1.027 times higher odds (**Table 3**), which was a logical variable to be significant given the direct link between blood glucose and diabetes. In comparison, insulin, which intuitively would be a predictor, wasn't significant and was removed from the final model. Insulin's lack of relevance is surprising and could be investigated further.

Table 3: Logistic regression coefficients, odds ratios, and 95% confidence intervals for the final model

Variable	Coefficient (β)	Odds Ratio	95% CI (Lower)	95% CI (Upper)
Glucose	0.036	1.037	1.027	1.047
Mass	0.074	1.077	1.036	1.122
Pedigree	1.087	2.966	1.327	6.871
Age	0.053	1.054	1.028	1.083

3 Classification Performance

3.1 Confusion Matrix and Performance Metrics

To test the accuracy of the model a confusion matrix was built. This will allow comparison of the models predictions compare to the real outcome. From this important metrics such as positive predictive value can be calculated. The predicted probabilities from the final model were calculated for all individuals in the dataset. A summary table for six individuals' predicted probability of diabetes and predicted class is shown in (**Table 4**). A confusion matrix was then created (**Table 5**). This showed the number of individuals correctly predicted to not have diabetes (n=233), the number correctly predicted to have diabetes (n=79), the number predicted to have diabetes that did not (n=29), and the number predicted to not have diabetes that did (n=51).

Table 4: Predicted probabilities and classification of diabetes status using a 0.5 threshold

Observation	$P(\textbf{diabetes} = 1)$	**Predicted Class**
4	0.02976	0
5	0.90981	1
7	0.03511	0
9	0.90799	1
14	0.92751	1
15	0.76457	1

Table 5: Confusion Matrix

Predicted	**Observed**	
	0	1
0	233	51
1	29	79

Table 6: Classification performance metrics for the logistic regression model using a probability threshold of 0.5

Metric	**Value (%)**
Sensitivity	0.608
Specificity	0.889
Positive predictive value	0.731
Negative predictive value	0.820
Accuracy	0.796

3.2 Clinical Interpretation

The accuracy of this model is fairly high at (79.6%) (**Table 6**). However, the sensitivity, which is the true positive rate, was only (60.8%). Meaning approximately 40% of people who did have diabetes were reported as negative. The specificity was high at (88.9%), meaning only approximately (12%) were incorrectly predicted to be positive (**Table 6**).

In this instance, the consequence of a false negative is more severe than a false positive. Untreated diabetes can lead to death [4]. Any positive in a predictive algorithm would likely result in that patient having a fasted blood glucose test, and only after clinical confirmation would medication be prescribed. False positives would likely be filtered out at this stage, thus the relative risk of being incorrectly treated is low. Based on this rationale, the threshold of this algorithm should be adjusted (e.g > 90%), sensitivity to encompass a greater percentage of the true positive cases.

4 ROC Curve and AUC Analysis

To visually examine the model accuracy, a receiver operator curve (ROC) was generated in (**Figure 2**). Visually, it appears that the furthest point on the curve slightly overshoots the top left corner. However, the area under the curve (AUC) will be calculated for a more objective measure. The AUC output (AUC = 0.8605 CI:0.82-0.90) can be seen in (**Table 7**). An AUC of 1 would be optimum, so the AUC of 0.86 indicates the model is relatively good at correctly classifying the data. This is in line with the accuracy reported in (**Table 6**). Though in future models, prioritising increased sensitivity over accuracy would be clinically pertinent.

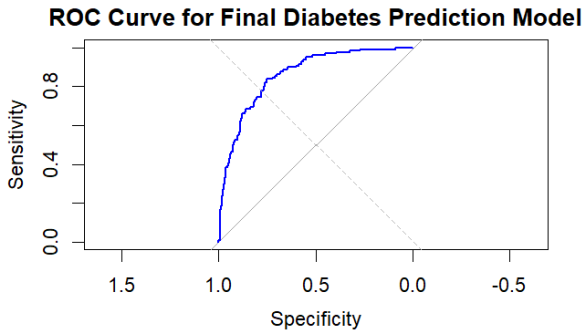


Figure 2: ROC curve for diabetes predictive model.

Table 7: AUC Performance of Logistic Regression Model

Metric	Value
Area Under the Curve (AUC)	0.8605
95% Confidence Interval	0.8231 – 0.8978
Method	DeLong

4.1 Youden's J Index

Youden's J Index calculates what cut-off threshold gives the optimal sensitivity specificity balance. The output is shown in (**Table 8**). The threshold previously used was 0.5, whereas Youden's Index reports the optimal threshold to be 0.265. Which would result in a sensitivity of 83.8% and a specificity of 75.2%.

Table 8: Optimal Threshold Based on Youden's J Index

Threshold	Sensitivity	Specificity
0.2650659	0.8384615	0.7519084

4.2 Model comparison using ROC's and AUC's

To compare the initial model, which contained all possible variables (model_1) and the reduced model with only significant variables (model_2), the ROC's were plotted on the same graph (**Figure 3**). There wasn't a significant visible difference between the two models. The AUC for both models was calculated, shown in (**Table 9**). The AUC was slightly higher for the full model (AUC = 0.86 CI: 0.83-0.89) than the reduced model (AUC= 0.82 CI: 0.82-0.90). This indicates that the model with all variables included is marginally better at correctly classifying the data type. Models lower in complexity tend to perform better on future data and the AIC was lower for model_2. In conclusion, model_2, with threshold modifications, is still the better choice.

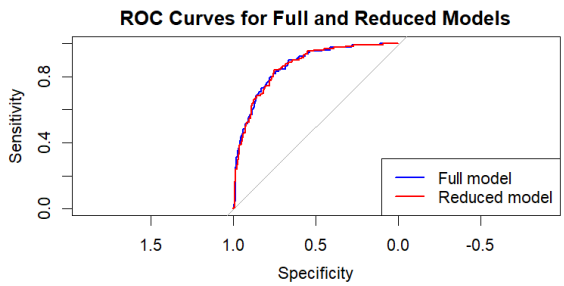


Figure 3: ROC curves for full and reduced models

Table 9: ROC AUC comparison of full and reduced logistic regression models (DeLong 95% CI).

Model	AUC	95% CI (DeLong)
Full model (`model_1`)	0.862	0.825–0.900
Reduced model (`model_2`)	0.861	0.823–0.898

5 Conclusion

In conclusion, the most significant predictors of diabetes in the Pima Indian women, using logistic regression, are plasma glucose concentration, BMI, the diabetes pedigree function and age in years. This does not account for interaction or squared terms, which can increase predictive accuracy through their ability to mathematically accommodate the fluctuating relationships between variables. However, despite its simplicity, the advantage of the logistic model over more accurate predictive models, such as Lasso, is that they provide a clearer basis for clinical interpretation of a given variable. Future work could investigate the biological basis for the lack of predictive value of insulin. Future work would also benefit from an updated dataset.

References

1. Kahn M. Diabetes. 1990. DOI: 10.24432/C5T59G. Available from: https://doi.org/10.24432/C5T59G
2. Fazeli PK, Lee H, and Steinhauser ML. Aging Is a Powerful Risk Factor for Type 2 Diabetes Mellitus Independent of Body Mass Index. eng. Gerontology 2020; 66:209–10. DOI: 10.1159/000501745
3. Ruze R, Liu T, Zou X, Song J, Chen Y, Xu R, Yin X, and Xu Q. Obesity and type 2 diabetes mellitus: connections in epidemiology, pathogenesis, and treatments. Frontiers in Endocrinology 2023 Apr; 14:1161521. DOI: 10.3389/fendo.2023.1161521. Available from: https://www.frontiersin.org/articles/10.3389/fendo.2023.1161521/full
4. Tomic D, Shaw JE, and Magliano DJ. The burden and risks of emerging complications of diabetes mellitus. en. Nature Reviews Endocrinology 2022 Sep; 18:525–39. DOI: 10.1038/s41574-022-00690-7. Available from: https://www.nature.com/articles/s41574-022-00690-7

A R Code

```
library(mlbench)
library(pROC)
library(ggplot2)
library(dplyr)
library(caret)
library(tidyverse)
library(tidyr)

if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
BiocManager::install("epiDisplay")
library(epiDisplay)

data("PimaIndiansDiabetes")
diab <- PimaIndiansDiabetes
view(diab)
class(diab)
str(diab)
names(diab)
head(diab)

# right lets get rid of implausible values
data("PimaIndiansDiabetes")
diabetes <- PimaIndiansDiabetes

cols <- c("glucose", "pressure", "triceps", "insulin", "mass", "pedigree",
    "age")
diabetes[cols] <- lapply(diabetes[cols], function(x) {
```

```
  x[x == 0] <- NA
  x
})
head(diabetes)

diab <- na.omit(diabetes)
diab

# now onto summary stats
# strat first
diab_neg <- diab[diab$diabetes == "neg", ]
diab_pos <- diab[diab$diabetes == "pos", ]

library(dplyr)

diab_summary <- diab %>%
  group_by(diabetes) %>%
  summarise(
    across(
      c(pregnant, glucose, pressure, triceps, insulin, mass, pedigree, age),
      list(
        Mean = ~mean(., na.rm = TRUE),
        SD   = ~sd(., na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )

View(diab_summary)

# now figs for most significant predictors
diab_long <- diab_summary %>% # into long format for gg
  pivot_longer(
    cols = -diabetes,
    names_to = c("variable", ".value"),
    names_pattern = "^(.*)_(Mean|SD)$"
  )

names(diab_long)

# Subset to glucose only
tmp <- diab_long[diab_long$variable == "glucose", ]

# Ensure correct order: neg then pos
tmp$diabetes <- factor(tmp$diabetes, levels = c("neg", "pos"))
tmp <- tmp[order(tmp$diabetes), ]

# Extract values
means <- tmp$Mean
sds   <- tmp$SD
```

```r
# Draw bar plot
bp <- barplot(
  means,
  names.arg = c("No␣diabetes", "Diabetes"),
  col = c("grey70", "grey40"),
  ylim = c(0, max(means + sds) * 1.15),
  ylab = "Mean␣glucose",
  main = "Glucose␣by␣diabetes␣status",
  border = "black"
)

# Add SD error bars
arrows(
  x0 = bp, y0 = means - sds,
  x1 = bp, y1 = means + sds,
  angle = 90,
  code = 3,
  length = 0.08,
  lwd = 2
)

tmp <- diab_long[diab_long$variable == "pregnant", ]
tmp$diabetes <- factor(tmp$diabetes, levels = c("neg", "pos"))
tmp <- tmp[order(tmp$diabetes), ]

means <- tmp$Mean
sds   <- tmp$SD

bp <- barplot(
  means,
  names.arg = c("No␣diabetes", "Diabetes"),
  col = c("grey70", "grey40"),
  ylim = c(0, max(means + sds) * 1.15),
  ylab = "Mean␣pregnancies",
  main = "Pregnancies␣by␣diabetes␣status",
  border = "black"
)

arrows(
  bp, means - sds,
  bp, means + sds,
  angle = 90, code = 3, length = 0.08, lwd = 2
)

tmp <- diab_long[diab_long$variable == "mass", ]
tmp$diabetes <- factor(tmp$diabetes, levels = c("neg", "pos"))
tmp <- tmp[order(tmp$diabetes), ]

means <- tmp$Mean
sds   <- tmp$SD
```

```
bp <- barplot(
  means,
  names.arg = c("No␣diabetes", "Diabetes"),
  col = c("grey70", "grey40"),
  ylim = c(0, max(means + sds) * 1.15),
  ylab = "Mean␣BMI",
  main = "BMI␣by␣diabetes␣status",
  border = "black"
)

arrows(
  bp, means - sds,
  bp, means + sds,
  angle = 90, code = 3, length = 0.08, lwd = 2
)

tmp <- diab_long[diab_long$variable == "pressure", ]
tmp$diabetes <- factor(tmp$diabetes, levels = c("neg", "pos"))
tmp <- tmp[order(tmp$diabetes), ]

means <- tmp$Mean
sds   <- tmp$SD

bp <- barplot(
  means,
  names.arg = c("No␣diabetes", "Diabetes"),
  col = c("grey70", "grey40"),
  ylim = c(0, max(means + sds) * 1.15),
  ylab = "Mean␣blood␣pressure",
  main = "Blood␣pressure␣by␣diabetes␣status",
  border = "black"
)

arrows(
  bp, means - sds,
  bp, means + sds,
  angle = 90, code = 3, length = 0.08, lwd = 2
)

tmp <- diab_long[diab_long$variable == "age", ]
tmp$diabetes <- factor(tmp$diabetes, levels = c("neg", "pos"))
tmp <- tmp[order(tmp$diabetes), ]

means <- tmp$Mean
sds   <- tmp$SD

bp <- barplot(
  means,
  names.arg = c("No␣diabetes", "Diabetes"),
  col = c("grey70", "grey40"),
  ylim = c(0, max(means + sds) * 1.15),
```

```
  ylab = "Mean␣age",
  main = "Age␣by␣diabetes␣status",
  border = "black"
)

arrows(
  bp, means - sds,
  bp, means + sds,
  angle = 90,
  code = 3,
  length = 0.08,
  lwd = 2
)

#### Task 2 ### make le model
diab$diabetes <- as.character(diab$diabetes)

# convert pos and neg to binary
diab$diabetes[diab$diabetes == "pos"] <- "1"
diab$diabetes[diab$diabetes == "neg"] <- "0"
diab$diabetes <- as.numeric(diab$diabetes)
table(diab$diabetes)
str(diab$diabetes)

# full logistic model with all predictors included
model_1 <- glm(
  diabetes ~ pregnant + glucose + pressure + triceps + insulin + mass +
    pedigree + age,
  data = diab,
  family = binomial
)
summary(model_1)

# reduced model with non-significant predictors removed
model_2 <- glm(
  diabetes ~ glucose + mass + pedigree + age,
  data = diab,
  family = binomial
)
summary(model_2)

# get info for table
coef_table <- summary(model_2)$coefficients
coef_table

OR <- exp(coef(model_2))
CI <- exp(confint(model_2))

results_table <- data.frame(
  Variable    = names(OR),
  Coefficient = coef(model_2),
```

```
  Odds_Ratio  = OR,
  CI_lower    = CI[, 1],
  CI_upper    = CI[, 2]
)

results_table
results_table[, -1] <- round(results_table[, -1], 3)
results_table

### Task 3 ###
diab_predicting <- predict(model_2, type = "response")
head(round(diab_predicting, 5))

diab_class <- ifelse(diab_predicting >= 0.5, 1, 0)
head(diab_class)

## making confusion matrix
conf_mat <- table(
  Predicted = diab_class,
  Observed  = model.frame(model_2)$diabetes
)
conf_mat

## sens, spes ect ##
TN <- conf_mat["0", "0"]
FN <- conf_mat["0", "1"]
FP <- conf_mat["1", "0"]
TP <- conf_mat["1", "1"]

sensitivity <- TP / (TP + FN)
specificity <- TN / (TN + FP)
ppv         <- TP / (TP + FP)
npv         <- TN / (TN + FN)
accuracy    <- (TP + TN) / sum(conf_mat)

round(
  c(
    Sensitivity = sensitivity,
    Specificity = specificity,
    PPV         = ppv,
    NPV         = npv,
    Accuracy    = accuracy
  ),
  3
)

## onto AUC and ROC
library(pROC)

roc_curve <- roc(
  response  = model.frame(model_2)$diabetes,
```

```
  predictor = diab_predicting
)

auc(roc_curve)
ci.auc(roc_curve)

plot(
  roc_curve,
  main = "ROC␣Curve␣for␣Final␣Diabetes␣Prediction␣Model",
  col = "blue",
  lwd = 2
)

abline(a = 0, b = 1, lty = 2, col = "grey")

### youdens
youden <- coords(
  roc_curve,
  x = "best",
  best.method = "youden",
  ret = c("threshold", "sensitivity", "specificity")
)
youden

### plotting model 1 (full) vs model2 (reduced with onnly signif var)
roc_full <- roc(
  response  = model.frame(model_1)$diabetes,
  predictor = predict(model_1, type = "response")
)

# Reduced model
roc_reduced <- roc(
  response  = model.frame(model_2)$diabetes,
  predictor = predict(model_2, type = "response")
)

plot(
  roc_full,
  col = "blue",
  lwd = 2,
  main = "ROC␣Curves␣for␣Full␣and␣Reduced␣Models"
)

plot(
  roc_reduced,
  col = "red",
  lwd = 2,
  add = TRUE
)

legend(
```

```
  "bottomright",
  legend = c("Full␣model", "Reduced␣model"),
  col = c("blue", "red"),
  lwd = 2
)

abline(a = 0, b = 1, lty = 2, col = "grey")

auc(roc_full)
ci.auc(roc_full)

auc(roc_reduced)
ci.auc(roc_reduced)

anova(model_1, model_2)
```